Diagonalization Continued, $\vec{\S}$5.3

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that

$$Av = \lambda v.$$

- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \ldots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \ldots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- A basis of $\mathbb{R}^n$ diagonalizing A is called an **eigenbasis**.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.

1. Let $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize A.

Theorem 6: An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.

2. Prove the theorem.

3. In each part, determine if there is a basis in which the linear transformation is represented by a diagonal matrix.

(a) Let $T : \mathbb{P}_2 \to \mathbb{P}_2$ given by $T(p) = p(t) + (t+1)p'(t)$.

(b) Let $T : \mathbb{P}_2 \to \mathbb{P}_2$ given by $T(p) = p(1) + p'(0)t + (p'(0) + p''(0))t^2$.

Diagonalization Again– Answers / Solutions

1. In order to find the eigenvalues, we compute the zeros of the characteristic polynomial. Note $p_A(\lambda) = \det \begin{bmatrix} 2-\lambda & 4 & 3 \\ -4 & -6-\lambda & -3 \\ 3 & 3 & 1-\lambda \end{bmatrix} = -(\lambda-1)(\lambda-2)^2$. Thus the eigenvalues of A are 1 and -2 (with algebraic multiplicity 2). We find the eigenspaces for each now.

$\underline{\lambda = 1}$. In this case, we get

$$E_1 = N(A - I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

$\underline{\lambda = -2}$. In this case, we get

$$E_{-2} = N(A + 2I_3) = \text{Span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Hence we have only two linearly independent eigenvectors and by the Diagonalization theorem, A cannot be diagonalized.

2. *Proof.* Suppose A is an $n \times n$ matrix with n distinct eigenvalues. Let $v_1, \ldots, v_n$ be n eigenvectors each corresponding to a different eigenvalue. We proved last time that eigenvectors corresponding to different eigenvalues must be linearly independent. Thus by the Diagonalization Theorem, A must be diagonalizable. $\square$

3. In each part we use the standard basis $\mathcal{B} = \{1, t, t^2\}$.

(a) Plugging in each basis vector to T, we get

$$T(1) = 1, T(t) = 2t + 1, T(t^2) = 3t^2 + 2t.$$

Expressing each in terms of basis $\mathcal{B}$, we get $[T(1)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $[T(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, and $[T(t^2)]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$.

Thus

$$[T]_{\mathcal{B}} = [[T(1)]_{\mathcal{B}}\ [T(t)]_{\mathcal{B}}\ [T(t^2)]_{\mathcal{B}}] = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Since $[T]_{\mathcal{B}}$ is upper-triangular, the eigenvalues are the diagonal entries. Thus the eigenvalues of T are 1, 2, and 3. We compute each eigenspace.

$\underline{\lambda = 1}$ Since $[T]_{\mathcal{B}} - I_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$, we get

$$N([T]_{\mathcal{B}} - I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Thus the eigenspace for $\lambda = 1$ is given by $E_1 = \text{Span}\{1\}$.

$\underline{\lambda = 2}$ Since $[T]_{\mathcal{B}} - 2I_3 = \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, we get

$$N([T]_{\mathcal{B}} - 2I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Thus the eigenspace for $\lambda = 2$ is given by $E_2 = \text{Span}\{1 + t\}$.

$\underline{\lambda = 3}$ Since $[T]_{\mathcal{B}} - 3I_3 = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$, we get

$$N([T]_{\mathcal{B}} - 3I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

Thus the eigenspace for $\lambda = 3$ is given by $E_3 = \text{Span}\{1 + 2t + t^2\}$.

Finally we take the basis $\mathcal{C} = \{1, 1 + t, 1 + 2t + t^2\}$, and then $[T]_{\mathcal{C}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

(b) Plugging in each basis vector to T, we get

$$T(1) = 1, T(t) = 1 + t + t^2, T(t^2) = 1 + 2t^2.$$

Expressing each in terms of basis $\mathcal{B}$, we get $[T(1)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $[T(t)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and $[T(t^2)]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

Thus

$$[T]_{\mathcal{B}} = [[T(1)]_{\mathcal{B}} \ [T(t)]_{\mathcal{B}} \ [T(t^2)]_{\mathcal{B}}] = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of A is $p_A(\lambda) = \det(A - \lambda I_n) = -(\lambda-1)^2(\lambda-2)$.. The eigenvalues of A are 1 (with algebraic multiplicity 2) and 2 (with algebraic multiplicity 1).

$\underline{\lambda = 1}$ Since $[T]_{\mathcal{B}} - I_3 = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, we get

$$N([T]_{\mathcal{B}} - I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Thus the eigenspace for $\lambda = 1$ is given by $E_1 = \text{Span}\{1, t - t^2\}$.

$\underline{\lambda = 2}$ Since $[T]_{\mathcal{B}} - 2I_3 = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, we get

$$N([T]_{\mathcal{B}} - 2I_3) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Thus the eigenspace for $\lambda = 2$ is given by $E_2 = \text{Span}\{1 + t^2\}$.

Finally we take the basis $\mathcal{C} = \{1, t - t^2, 1 + t^2\}$, and then $[T]_{\mathcal{C}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.